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MedFMIT: A Foundation Model-Driven Multimodal Fusion Method for Image-Text Disease Diagnosis

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***Abstract*— Image-text multimodal disease diagnostic models have the potential to provide more precise diagnosis results compared to conventional image-only diagnostic models. Existing image-text multimodal diagnostic methods struggle to address the significant image-text distribution differences and realize comprehensive information interaction between the two modalities. To tackle these issues, we propose a novel foundation model-driven multimodal fusion model, MedFMIT, for image-text disease diagnosis. MedFMIT utilizes the DCA encoders to extract informative and well-aligned visual and textual feature representations, effectively reducing the distributional gap between images and texts while ensuring robust feature extraction. The DMII module is introduced to facilitate comprehensive information interaction between image and text features at both coarse-grained and fine-grained levels. For performance evaluation, we conducted experiments on two multimodal medical classification datasets (image + text) containing computed tomography images and endoscopic optical images. MedFMIT outperforms other state-of-the-art multimodal algorithms, achieving the AUC scores of 92.7% and 85.4% on the two datasets respectively, demonstrating its strong potential for precise medical diagnosis.**

***Keywords*— Multi-modal fusion, Medical texts, Foundation model, Vision-language model, Disease diagnosis**

I. Introduction

With the rapid development of deep learning in recent years, computer-aided diagnosis has gained significant attention due to its non-invasive and convenient nature [1]. However, most existing disease diagnosis methods rely solely on medical images. Due to the challenges of medical data collection and the difficulties of data sharing, medical datasets are often small [2] (typically containing fewer than 1,000 samples). This limitation in dataset size significantly impacts the performance of image-only diagnostic models, as deep learning models generally require large amounts of data to effectively learn complex patterns. Therefore, effectively integrating additional information, such as medical texts accompanying images, has become an urgent need for training more accurate diagnostic models. Medical texts often provide detailed descriptions of lesions, such as locations, shapes, sizes, and other characteristics. These texts not only emphasize but also complement the information contained within the images, offering the potential to further enhance diagnostic performance.

The combined use of medical images and text in deep learning has been explored in several studies. Models such as MedCLIP [3] and BiomedCLIP [4] leverage large-scale medical image-text pairs to train foundation vision-language models (VLM), enabling effective cross-modal alignment between images and text. The success of medical VLM has inspired the integration of medical texts in image segmentation tasks. Methods like LViT [5], TGCAM [6], and MMI-UNet [7] incorporate medical text into segmentation networks through techniques like attention mechanisms, improving the performance of medical image segmentation.

Despite these advancements, the development of medical image-text classification algorithms remains relatively underexplored. Early fusion [8] and late fusion [9] are two commonly used approaches for multimodal classification. They combine image and text features through straightforward concatenation or averaging, achieving basic feature integration. Irene [10] employs bidirectional multimodal attention blocks and self-attention blocks to jointly process medical images and unstructured chief complaints, facilitating information interaction to some extent.

While existing image-text classification methods demonstrate strong performance, there is still much room for improvement in two key aspects. The first aspect is achieving effective bimodal feature alignment. Medical images and texts represent two distinct forms of information with significantly different feature distributions. Misalignment of image and text features may damage the quality of feature fusion. The second aspect is the comprehensive utilization of both coarse-grained and fine-grained information in two modalities. Diagnostic information in images and texts can range from specific local details to broader overall representations. Fully exploiting both fine-grained and coarse-grained information during the bimodal fusion stage can enhance the interaction between modalities, enabling the extraction of more informative and accurate diagnostic insights.

Here, we propose MedFMIT, a **F**oundation model-driven **M**ultimodal fusion method for **I**mage-**T**ext **Med**ical diagnosis. Our contributions can be summarized as follows:

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1. We introduce a multimodal fusion network based on a foundation VLM for joint diagnosis of medical images and medical texts.
2. We incorporate domain-specific cross-modal alignment (DCA) encoders into our network, which not only have powerful feature extraction capabilities but also achieve effective alignment between visual and textual features, laying a foundation for subsequent multimodal fusion.
3. We propose a dual-branch multi-granularity intermodal interaction (DMII) module to fuse information from medical images and texts, which enables comprehensive cross-modal interaction using both coarse-grained and fine-grained feature representations, further enhancing disease diagnosis performance.
4. Experimental results on two medical datasets demonstrate that our MedFMIT model achieves SOTA performance, with an AUC of 92.7% for COVID-19 classification and 85.4% for distinguishing colorectal polyp neoplasms. These results highlight the potential of MedFMIT for accurate disease diagnosis.

II. Method

A. Overview of MedFMIT

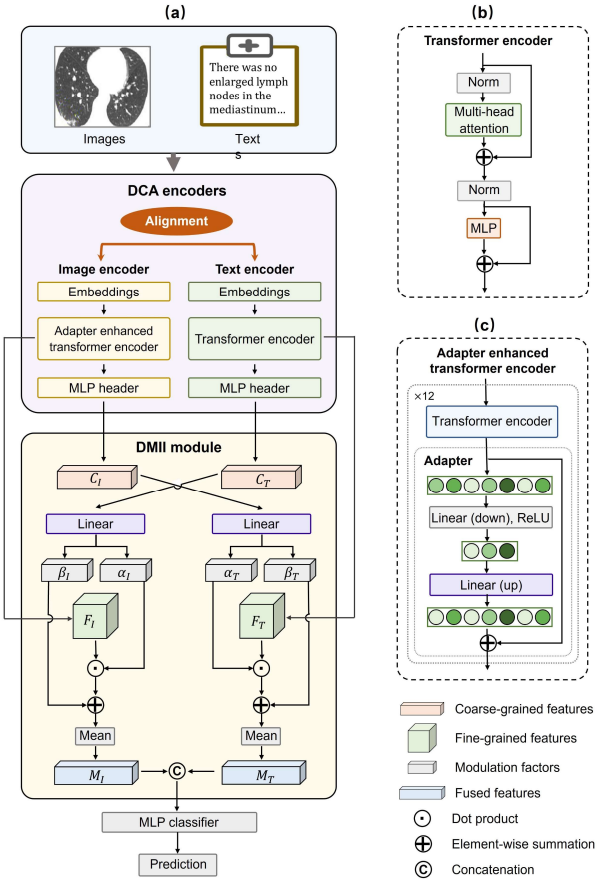


Fig. 1. The overview of our proposed model. (a) The architecture of MedFMIT. (b) The detailed structure of the transformer encoder. (c) The detailed structure of the adapter enhanced transformer encoder.

The architecture of MedFMIT is depicted in Fig. 1. Medical images and texts are fed into the foundation model-based DCA encoders to extract both fine-grained and coarse-grained features. These visual and textual features are then integrated through the DMII module, facilitating multimodal interaction. The fused features are subsequently passed through a multi-layer perceptron (MLP) classifier, producing the final diagnosis prediction.

B. Domain-specific Feature Alignment with DCA Encoders

We introduce the DCA encoders to extract insightful and well-aligned information from both images and texts. BiomedCLIP is a popular medical foundation VLM, which exhibits remarkable proficiency in constructing medical image-text embedding spaces. Leveraging a contrastive learning objective, BiomedCLIP effectively aligns image and text features. We efficiently finetune the domain-specific BiomedCLIP model for our diagnosis tasks, transforming it into a powerful feature extractor. Unlike most conventional multimodal classification tasks, the visual and textual features extracted through this method are not isolated from each other, but inherently interconnected, laying the foundation for subsequent bimodal fusion.

Adapter-enhanced image encoders. To enhance the adaptability of BiomedCLIP's image encoder for specific diagnostic tasks while preserving its robust medical image feature extraction capabilities, we incorporate adapter blocks into the frozen foundation image encoder to capture task-specific image domain characteristics. The medical image is first preprocessed with patch and position embeddings, then passed through the adapter-enhanced transformer encoder to generate a feature representation $F_I \in R^{B\times196\times768}$, where B refers to batch size, 196 represents the number of patches, and 768 denotes the feature dimension. F_I captures patch-level contextual features, which can be interpreted as a kind of fine-grained feature representation. This representation is further refined by an MLP header to produce a more abstract feature representation $C_I \in R^{B\times512}$, where 512 is the feature dimension. C_I captures the global information of the medical image, which is a kind of coarse-grained feature representation. In the adapter-enhanced transformer encoder, an adapter block is inserted after each transformer encoder to finetune the model. The adapter block is a lightweight bottleneck structure consisting of a down-projection and up-projection linear layer.

Text encoders. The medical text is first preprocessed by applying word embeddings and position embeddings, and then fed into the transformer encoder to obtain a feature representation $F_T \in R^{B\times256\times768}$, where 256 is the number of tokens and 768 is the feature dimension. F_T captures the token-level contextual features, serving as fine-grained textual information. Subsequently, F_T is further refined through an MLP header to generate $C_T \in R^{B\times512}$, which represents the coarse-grained information of the medical texts. The text encoder utilizes the pre-trained weights of BiomedCLIP, which remain frozen during training.

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C. Multi-granularity Feature Fusion with DMII Module

The well-aligned visual features and textual features obtained from the DCA encoders are further fed into the DMII module to realize multi-granularity feature fusion. The DMII module consists of parallel dual branches. In the left branch, we use the coarse-grained text representation C_T to modulate the fine-grained image representation F_I. Inspired by [11], we employ a simple linear layer to generate the modulate factors $\alpha_I \in R^{B\times 196}$ and $\beta_I \in R^{B\times 196}$, where 196 is the number of patches. The modulate factors are then applied to F_I to modulate it at the patch-level dimension (the second dimension of F_I):

$$F_I' = \alpha_I \cdot F_I + \beta_I \tag{1}$$

This type of modulation leverages text information to further highlight and enrich image information. Subsequently, F_I' is averaged along the second dimension to obtain the text-modulated image features M_I.

The structure of the right branch is similar to that of the left branch, where we use the coarse-grained image representation C_I to modulate the fine-grained text representation F_T and generate the image-modulated text features M_T. Finally, we concatenate M_I and M_T for the final diagnosis prediction.

D. Loss Function

We adopt cross-entropy loss as the loss function, which is defined as:

$$L = -\sum_{i=1}^{C} y_i \log(p_i) \tag{2}$$

where C is the total number of classes. y_i and p_i represent the ground truth and the predicted probability for the i-th class.

III. Experiments and Results

A. Datasets

We utilized two medical datasets containing both images and corresponding texts for performance evaluation. The first dataset is the COV-CTR [12] dataset, which consists of 726 image-text pairs for COVID-19 diagnosis. In the COV-CTR dataset, the medical images are computed tomography images and the medical texts are medical reports for corresponding images. The second dataset is the BKAI dataset [13], which is used to distinguish between neoplastic colorectal polyps and non-neoplastic colorectal polyps. The medical images in the BKAI dataset are endoscopic optical images, and the medical texts are the corresponding textual descriptions of these images generated by [14]. We filtered the images that contained both neoplastic and non-neoplastic polyps, resulting in a total of 951 samples. For both datasets, we fixed 20% of the samples as the test set and randomly split the remaining samples into training and validation sets in a 4:1 ratio to conduct 5-fold cross-validation. A summary of these two datasets is illustrated in Table I. The example medical images and their corresponding medical texts from the two datasets are shown in Fig.2.

TABLE I. Summary of Two Datasets

Datasets	Classes	
COV-CTR Dataset	**Non-COVID-19**	**COVID-19**
	391	335
BKAI Dataset	**Non-neoplastic polyps**	**Neoplastic polyps**
	257	694

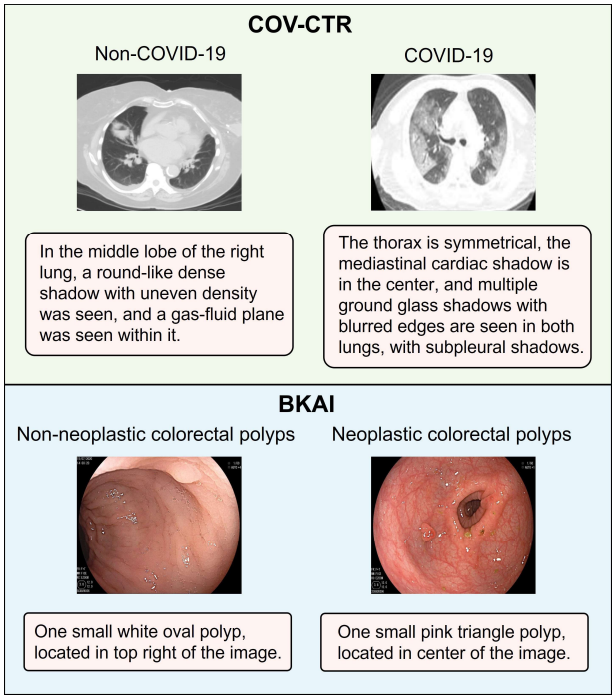


Fig. 2. Example medical images and their corresponding medical texts from the two datasets.

B. Implementation Details

During the training process of MedFMIT, we set the training epochs to 100. The batch size was set to 32, and the optimizer was AdamW. The learning rate was set as 5e-3 for the COV-CTR dataset and 1e-3 for the BKAI dataset. The compared methods include two multimodal models designed for natural images (ViLT [15] and GIT [16]), and one image-text fusion method for clinical diagnostics (Irene [10]). The resolutions of input images were adjusted to 224 × 224 for all the compared algorithms. To address the challenge of data imbalance in the BKAI dataset, all models for the BKAI dataset adopted a weighted sampling strategy during the training phase. All experiments were conducted using the Pytorch framework, using one NVIDIA Tesla A800 GPU for training. The workstation is Precision 7920 Tower, equipped with an Intel (R) Xeon (R) Silver 4210 R CPU@2.40 GHz.

C. Evaluation Metrics

To evaluate the performance of disease diagnosis, we use metrics including area under the curve (AUC), balanced accuracy (bACC), and Matthews correlation coefficient (MCC). AUC refers to the area under the Receiver Operating

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Characteristic curve, which plots the true positive rate against the false positive rate across various threshold settings of a classifier. A higher AUC value indicates better overall classifier performance bACC is a commonly used metric for classification, which is defined as:

$$bACC = \frac{\frac{TP}{TP+FN} + \frac{TN}{TN+FP}}{2} \quad (3)$$

where the number of true positive (TP), false positive (FP), false negative (FN), and true negative (TN) can be obtained from the confusion matrix. Higher bACC indicates better performance. MCC is generally regarded as a balanced measurement because it comprehensively takes TP, FP, FN, and TN into account, which is defined as:

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \quad (4)$$

A higher MCC indicates better classification performance.

D. Comparison with SOTA Image-text Fusion Methods

We compare MedFMIT with two multimodal models designed for natural images (ViLT and GIT) and one image-text fusion method tailored for clinical diagnostics (Irene). The performance metrics for all methods are assessed using 5-fold cross-validation and are reported as mean ± standard deviation in Table II. Additionally, the metrics for all five folds are illustrated using boxplots in Fig. 3. MedFMIT exhibits superior diagnostic capabilities compared to other algorithms, achieving AUC scores of 92.7% and 85.4% on the two datasets respectively.

TABLE II. COMPARISONS WITH OTHER MULTIMODAL METHODS

Model	Classification performance		
	AUC	bACC	MCC
	COV-CTR dataset		
ViLT	0.914±0.017	0.855±0.017	0.711±0.033
GIT	0.848±0.011	0.808±0.008	0.619±0.016
Irene	0.897±0.007	0.819±0.012	0.638±0.021
Ours	**0.927±0.004**	**0.887±0.019**	**0.770±0.037**
	BKAI dataset		
ViLT	0.796±0.047	0.743±0.028	0.468±0.067
GIT	0.830±0.045	0.737±0.035	0.463±0.081
Irene	0.827±0.036	0.744±0.041	0.479±0.074
Ours	**0.854±0.006**	**0.783±0.011**	**0.514±0.017**

MedFMIT outperforms ViLT and GIT, which simply concatenate the extracted image and text features before feeding them into transformer layers for bimodal fusion. Specifically, compared with ViLT and GIT, MedFMIT achieves substantial improvements of 8% and 24% in MCC on the COV-CTR dataset. Similarly, on the BKAI dataset, MedFMIT demonstrates significant gains of 10% and 11% in MCC. These improvements can be attributed to MedFMIT's ability to achieve better feature alignment and fusion between the two modalities through its DCA encoders and DMII module. In contrast, the bimodal interaction in ViLT and GIT is limited due to their simple concatenation strategy.

When compared to Irene, which employs bidirectional multimodal attention blocks and self-attention blocks to enhance the interaction between visual and textual features, MedFMIT also achieves superior performance. MedFMIT improves the bACC scores from 0.819 to 0.887 on the COV-CTR dataset and from 0.744 to 0.783 on the BKAI dataset. While Irene enables modality integration, MedFMIT's strong feature alignment capabilities of DCA encoders result in more effective information fusion. Furthermore, Irene's large parameter size makes it prone to overfitting on small datasets, a common challenge in medical diagnostic tasks. In contrast, MedFMIT leverages a fine-tuning strategy and incorporates the lightweight DMII module, enabling efficient training on small datasets and improving overall diagnostic performance.

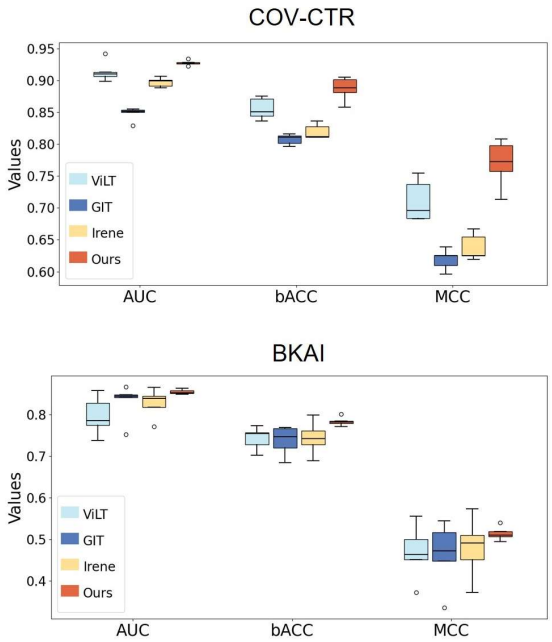


Fig. 3. Boxplots of 5-fold cross-validation performance for different multimodal fusion methods.

E. Ablation Studies

Image-text joint prediction. In the first ablation experiment, we aim to demonstrate the effectiveness of image-text joint prediction. We evaluate the classification performance of models using images, texts, and both images and texts. Image encoders and text encoders used in the image-only and text-only models are the same as those in MedFMIT.

The comparison results are presented in Table III and Fig. 4. Compared to the unimodal models, MedFMIT, which leverages both medical images and texts, achieves superior diagnostic performance. Specifically, the multimodal model achieves an improvement in MCC of 25% and 26% on the COV-CTR dataset, and 22% and 53% on the BKAI dataset. These results highlight the effectiveness of multimodal fusion in enhancing diagnostic performance.

TABLE III. ABLATION STUDY FOR MODALITY SELECTION

Model	Classification performance		
	AUC	bACC	MCC

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	COV-CTR dataset		
Image-only	0.870±0.013	0.809±0.019	0.614±0.037
Text-only	0.883±0.031	0.801±0.014	0.609±0.031
Image + Text	**0.927±0.004**	**0.887±0.019**	**0.770±0.037**
	BKAI dataset		
Image-only	0.802±0.013	0.734±0.016	0.423±0.029
Text-only	0.763±0.015	0.684±0.015	0.335±0.030
Image + Text	**0.854±0.006**	**0.783±0.011**	**0.514±0.017**

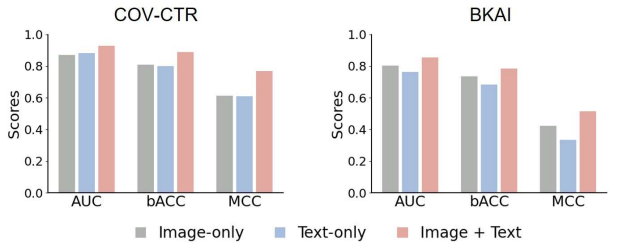


Fig. 4. Bar charts of ablation study results for modality selection.

Furthermore, we utilized the uniform manifold approximation and projection (UMAP) algorithm [17] to visualize the feature representation of models using images, texts, and a combination of both images and texts. The visualization results of two datasets are presented in Fig. 5. It can be observed that the image-text fusion models exhibit improved feature separation across different classification classes compared to models using only images or texts, highlighting the advantages of integrating images and texts for disease diagnosis.

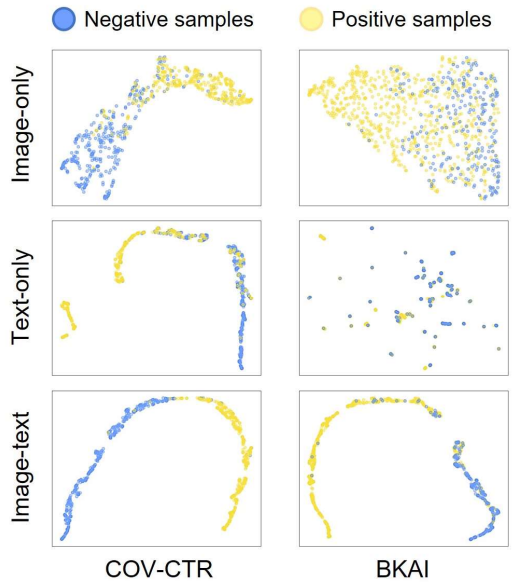


Fig. 5. The UMAP visualization of models using images, texts, and a combination of both images and texts.

Feature extraction — DCA encoders. In the second ablation experiment, we try to prove the superiority of DCA encoders. We hypothesize that leveraging the pre-trained weights of the foundation VLM BiomedCLIP would be beneficial for extracting insightful and well-aligned visual and textual features. To validate this hypothesis, we replace the pre-trained weights of the image branch with the ViT-base model's pre-trained weights on ImageNet, and replaced the weights of the text branch with the BERT-base-uncased model's weights pre-trained on BooksCorpus and English Wikipedia. These replacements not only result in the loss of BiomedCLIP's domain-specific feature extraction capabilities but also disrupt the alignment between the image and text embeddings.

The comparison results are shown in Table IV. The symbol √ indicates the use of BiomedCLIP's pre-trained weights and the symbol × indicates the use of alternative pre-trained weights. As shown in Table IV, the DCA encoders utilizing BiomedCLIP's pre-trained weights for both the image and text encoders achieve the best and most robust classification performance，characterized by the highest mean and lowest standard deviation in AUC, bACC, and MCC. Specifically, when using BiomedCLIP's pre-trained weights for both encoders, the model achieves average MCC scores of 0.770 and 0.514 on the two datasets respectively. In contrast, models that do not utilize BiomedCLIP's pre-trained weights for either the image or text encoders generally exhibit the weakest performance, with average MCC scores of 0.557 and 0.415 on the two datasets respectively. The above quantitative results demonstrate that DCA encoders can extract more informative image-text features, thereby significantly improving diagnostic performance.

TABLE IV. ABLATION STUDY FOR DIFFERENT FEATURE ENCODERS

Pretrained		**Classification performance**		
Image	**Text**	**AUC**	**bACC**	**MCC**
		COV-CTR dataset		
×	×	0.839±0.060	0.779±0.066	0.557±0.129
√	×	0.904±0.025	0.843±0.040	0.683±0.081
×	√	0.858±0.050	0.795±0.056	0.591±0.111
√	√	**0.927±0.004**	**0.887±0.019**	**0.770±0.037**
		BKAI dataset		
×	×	0.819±0.008	0.727±0.021	0.415±0.047
√	×	0.835±0.011	0.734±0.029	0.447±0.036
×	√	0.822±0.019	0.725±0.019	0.416±0.038
√	√	**0.854±0.006**	**0.783±0.011**	**0.514±0.017**

Feature fusion — DMII module. In the third ablation experiment, we illustrate the advantages of the DMII module. We keep the DCA encoders unchanged and replace the fusion method with early fusion and late fusion. For early fusion, the image and text features extracted by the DCA encoders are concatenated and fed into an MLP classifier to generate the final prediction. For late fusion, the extracted image and text features are processed separately by two MLP classifiers to produce individual predictions, which are then averaged to obtain the final prediction.

The corresponding quantitative results are provided in Table V and Fig. 6. The DMII module achieves the MCC scores of 0.770 and 0.514 in two datasets, more than 8% higher than early fusion and late fusion. These results indicate that by enabling the image-text interaction from both coarse-grained level and fine-grained-level, the DMII module has learned the disease diagnosis information comprehensively.

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TABLE V. ABLATION STUDY FOR DIFFERENT FUSION METHODS

Fusion method	Classification performance		
	AUC	bACC	MCC
	COV-CTR dataset		
Early fusion	0.886±0.021	0.811±0.011	0.619±0.022
Late fusion	0.920±0.015	0.839±0.018	0.676±0.036
DMII	**0.927±0.004**	**0.887±0.019**	**0.770±0.037**
	BKAI dataset		
Early fusion	0.798±0.013	0.715±0.019	0.390±0.031
Late fusion	0.841±0.011	0.751±0.026	0.473±0.037
DMII	**0.854±0.006**	**0.783±0.011**	**0.514±0.017**

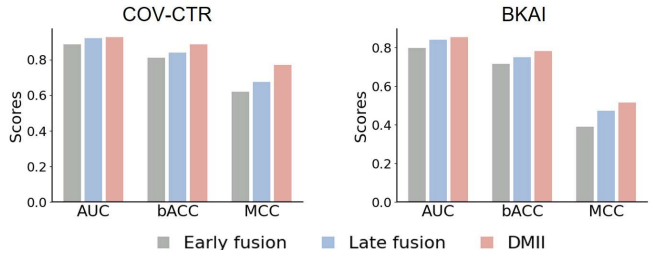


Fig. 6. Bar charts of ablation study results for different fusion methods.

IV. DISCUSSION AND CONCLUSION

This paper proposes a novel foundation model-based multimodal fusion method MedFMIT for disease diagnosis based on both medical images and texts. Unlike most fusion methods that simply feed independent image and text features to the fusion block, MedFMIT employs domain-specific cross-modal alignment encoders to extract informative and well-aligned image and text features, facilitating subsequent feature fusion. A multimodal fusion module DMII is then used to fuse the well-aligned image and text features, enabling multi-granularity intermodal interactions and further improving diagnostic performance. This is another improvement compared to other existing fusion methods.

Notably, the image branch of our model does not impose specific requirements on the image modality, which allows our method to be applied to various types of image-text datasets. The strong generalization capability highlights the broad clinical application prospects of our approach. Additionally, although a large foundation model is utilized in our method, we only fine-tune the foundation model during training, rather than performing full parameter optimization. The number of trainable parameters in our model is 4,217,271, and the GPU memory usage during training is approximately 4 GB, making it compatible with standard GPUs. Additionally, the average inference time per sample is only 0.012 seconds. These low GPU requirements and fast inference speeds provide a solid foundation for potential clinical applications.

Our method also has some limitations. Our model lacks validation by medical experts or deployment in real-world clinical settings. In the future, we plan to collaborate with clinical experts to further refine our method and explore its potential for clinical applications. Additionally, the texts used in the BKAI dataset are synthetic texts. While our experiments demonstrate that incorporating such generated text improved the diagnostic performance of the model, the reliability of generated text warrants careful consideration. Moving forward, we aim to develop evaluation metrics to assess the reliability of generated texts and to mitigate potential diagnostic risks associated with their use.

Overall, MedFMIT is a powerful multimodal fusion method for disease diagnosis. We hope that our MedFMIT can be beneficial for various types of computer-aided diagnosis tasks and promote the development of medical care.

STATEMENT

This study used public datasets and did not involve human participants or animal experiments, and thus no ethical approval was required.

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